

Statistics Homework 3:

Simple Contrasts

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Acceptable Abbreviations Page

This homework will ask you to indicate names of statistics by typing abbreviations from the Acceptable Abbreviations Page. The Acceptable Abbreviations Page is a document posted on Canvas.

Part A: Expressive writing studies

Denise Sloan and her colleagues (2008) tested the extent to which an expressive writing exercise reduced negative feelings in a sample of 68 college students. Participants were randomly assigned into two different groups, with each group completing a different daily, 20-minute writing session for three consecutive days. Group 1 was an expressive writing treatment condition in which participants were asked to write about the most stressful or traumatic experience of their lives with as much emotion and feeling as possible. Group 0 was a control condition in which participants were asked to write about how they spent their time each day without any emotion or opinions. A few weeks after completing the writing exercise, all participants rated their level of negative feelings using a 21-item scale called the Depression and Anxiety Symptom Scale (DASS). A higher score on this scale indicates having higher levels of depressed mood and anxious feelings.

Results similar to the results from this study are printed below. Use these results to answer the questions that follow.

Group Statistics

Group	N	Mean	Std. Deviation	Std. Error Mean
DASS 1.00	35	5.20	6.05	1.02
.00	33	8.70	7.63	1.33

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
DASS	Equal variances assumed	2.17	.145	-2.10		.039			-6.83	-.18
	Equal variances not assumed			-2.09		.041			-6.85	-.15

1. Compute the pooled variance
2. Compute the mean difference
3. Compute the standard error of difference
4. Compute the pooled standard deviation
5. Compute the degrees of freedom
6. Compute the point-biserial correlation
7. In the treatment group (Group 1), what is the distance that people's scores typically deviated from the group mean?
8. If this study were repeated over and over, how much would you typically expect the mean for the **treatment group** (Group 1) to deviate from the true population mean for this group?

9. Calculate the t -value that could be used to test whether the mean for the **treatment group** (Group 1) is significantly different from zero.
10. If the true population mean for the **treatment group** (Group 1) is zero, what is the probability of getting a sample mean that is as extreme as the mean obtained in this study?
11. If this study were repeated over and over, how much would you typically expect the size of difference between the two groups to deviate from the true difference in the population?
12. If the true population difference between groups is zero, what is the probability of getting a difference between groups that is as extreme as the difference obtained in this study? **(report exactly as it appears with no extra zeros and no rounding)**
13. If the null hypothesis about the difference between groups is true, what is the standardized $e_{1-0_{sp}}$ for this study?

Is there a difference between groups? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

14. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
 15. the value of the descriptive statistic
 16. the name of the inferential statistic (use the Acceptable Abbreviations Page)
 17. the value of the inferential statistic
 18. the p-value **(report exactly as it appears with no extra zeros and no rounding)**
19. This effect is best described as:
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

Danielle Arigo and her colleagues (2012) conducted a similar study investigating the effectiveness of an expressive writing exercise. This study used a sample of 111 college women, with 57 assigned to the treatment condition (Group 1), and 54 assigned to the control condition (Group 0). The follow-up assessment was completed 8 weeks after the writing exercises, and it included a measure of sleep problems called the Pittsburgh Sleep Quality Index. A higher score on this instrument indicates having greater difficulties with sleep.

Results similar to the results from this study are printed below. Use these results to answer the questions that follow.

Group Statistics

Group	N	Mean	Std. Deviation	Std. Error Mean
SleepProb 1.00	57	3.23	2.56	
.00	54	4.56	3.22	.44

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
SleepProb	Equal variances assumed	3.23	.07		109		-1.33	.55		
	Equal variances not assumed				101.03		-1.33	.55		

20. In this example, the pooled variance is 8.41. Compute the pooled standard deviation.
21. Compute the standard error of the mean for the treatment group (Group 1).
22. Compute the lower end of the 95% confidence interval.
23. Compute the upper end of the 95% confidence interval.
24. What is the average deviation of $|e_{1-0.sp}|$?

Is there a difference between groups? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

25. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
 26. the value of the descriptive statistic
 27. the name of the inferential statistic (use the Acceptable Abbreviations Page)
 28. the value of the inferential statistic
 29. the p-value
30. This effect is best described as:
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

Part B: What's a point-biserial correlation for?

The next set of questions will require you to identify what types of questions can be answered with a point biserial correlation and which questions cannot be answered with a point biserial correlation. Answer each question with one of the following options: (A) yes, (B) no, or (C) the given information does not address this issue.

You calculate a point biserial correlation, and the correlation is significant. Based on this information:

31. Was the t-test for mean differences significant?
32. Is there a difference between means?
33. Is there a difference between standard deviations?
34. Were any assumptions violated?
35. Does the rank order of cases show stability across different assessment points?

Part C: Motivational intervention to reduce college student drinking

Mark Prince and his colleagues (2015) developed a motivational intervention intended to reduce problematic drinking in college students. In this intervention, students first completed a questionnaire about their own drinking attitudes and behavior and then they received accurate feedback on their scores and how their scores compare to other students at their university. This feedback was provided in a 30-minute, individual meeting with a graduate student. It was expected that, after completing this intervention, students would lower their perceived "descriptive norms" for how much alcohol they think other students at their university typically drink. They were also expected to drink less and thereby experience fewer negative consequences from drinking. Prince conducted a study using 70 students, all of whom reported having at least one heavy drinking episode within the last month. Students were randomly assigned into two groups. Group 1 received the intervention, and Group 0 was a control group that completed all the same assessments as the treatment group but did not receive any feedback and did not meet with a graduate student. One month after the initial assessment and treatment, students completed an on-line follow-up questionnaire that included a measure alcohol related consequences (higher scores indicated more experiences with alcohol related problems within the last month such as hangovers and memory blackouts), and a measure of descriptive norms (higher scores indicated higher levels of perceived peer drinking).

Results regarding the alcohol related consequences (ARC) outcome variable are printed below. Use these results to answer the questions that follow.

Group Statistics

group	N	Mean	Std. Deviation	Std. Error Mean
ARC 1.00	31	4.64	4.01	.72
.00	39	6.45	5.08	.81

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
ARC	Equal variances assumed	1.02	.32	-1.62		.11			-4.04	.42
	Equal variances not assumed			-1.66		.10			-3.98	.36

36. Compute the pooled variance
37. Compute the mean difference
38. Compute the standard error of difference
39. Compute the pooled standard deviation
40. Compute the degrees of freedom
41. Compute the point-biserial correlation
42. In the treatment group (Group 1), what is the distance that people's scores typically deviated from the group mean?
43. If this study were repeated over and over, how much would you typically expect the mean for the **treatment group** (Group 1) to deviate from the true population mean for this group?
44. Calculate the *t*-value that could be used to test whether the mean for the **treatment group** (Group 1) is significantly different from zero.
45. If the true population mean for the **treatment group** (Group 1) is zero, what is the probability of getting a sample mean that is as extreme as the mean obtained in this study?
46. If this study were repeated over and over, how much would you typically expect the size of difference between the two groups to deviate from the true difference in the population?
47. If the true population difference between groups is zero, what is the probability of getting a difference between groups that is as extreme as the difference obtained in this study? **(report exactly as it appears with no extra zeros and no rounding)**
48. If the null hypothesis about the difference between groups is true, what is the standardized $e_{1-0\ sp}$ for this study?

Is there a difference between groups? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

49. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
50. the value of the descriptive statistic
51. the name of the inferential statistic (use the Acceptable Abbreviations Page)
52. the value of the inferential statistic
53. the p-value **(report exactly as it appears with no extra zeros and no rounding)**
54. This effect is best described as:
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large



Results regarding the descriptive norms (DN) outcome variable are printed below. Use these results to answer the questions that follow.

Group Statistics

group	N	Mean	Std. Deviation	Std. Error Mean
DN 1.00	31	15.22	12.60	
.00	39	26.13	11.95	1.91

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference		Lower	Upper
DN	Equal variances assumed	.012	.912		68		-10.91	2.95			
	Equal variances not assumed				62.87		-10.91	2.96			

55. In this example, the pooled variance is 149.84. Compute the pooled standard deviation.

56. Compute the standard error of the mean for the treatment group (Group 1).

57. Compute the lower end of the 95% confidence interval.

58. Compute the upper end of the 95% confidence interval.

59. What is the average deviation of $|e_{1-0\ sp}|$?

Is there a difference between groups? To answer this question, provide the 5 pieces of information.

When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

60. the name of the descriptive statistic (use the Acceptable Abbreviations Page)

61. the value of the descriptive statistic

62. the name of the inferential statistic (use the Acceptable Abbreviations Page)

63. the value of the inferential statistic

64. the p-value

65. This effect is best described as:

(A) not significant (B) significant but trivial (C) small (D) medium (E) large

Part D: Transition to Parenthood

When a married couple has their first baby, is the new child likely to be beneficial or detrimental to the overall quality of their relationship? Feeney, Alexander, Noller, and Hohaus (2003) conducted a study investigating ways in which marriages change during the transition to parenthood. These researchers followed a sample 76 couples over the course of approximately one year. At the beginning of the study, all couples were expecting their first child. Participating couples completed one assessment during the second trimester of pregnancy (time 1), and they completed another assessment six months after the birth of their baby (time 2). As part of each assessment, each spouse completed a measure of marital satisfaction. A paired samples *t*-test can be used to determine if couples' marital satisfaction at time 2 (after the birth of the baby) is lower than their satisfaction at time 1 (before the birth of the baby).

Results from paired samples *t*-tests for wives and husbands are reported below.

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	w_sat_t1	31.77	76	4.50	.52
	w_sat_t2	29.73	76	5.15	.59

Results for wives' satisfaction

t1 = time 1
t2 = time 2

Paired Samples Correlations

	N	Correlation	Sig.
Pair 1 w_sat_t1 & w_sat_t2	76	.60	.000

Paired Samples Test

	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Pair 1 w_sat_t1 - w_sat_t2		4.33		1.06	3.03	4.12		.000

66. Compute the mean difference
67. Compute the standard error of mean difference
68. Compute the pooled variance
69. Compute the degrees of freedom
70. At time 2, what is the distance that people's scores typically deviated from the group mean?
71. If this study were repeated over and over, how much would you typically expect the mean for women at time 2 to deviate from the true population mean for women after the birth of a child?
72. What is the average difference score (that is, the difference between a person's score at time 1 and her score at time 2)?
73. What is the typical distance that people's difference scores deviated from the group mean difference score?
74. If this study were repeated over and over, how much would you typically expect the size of group mean difference score to deviate from the true difference in the population?
75. If the null hypothesis about the difference scores is true, what is the standardized $e_{diff\ sp}$ for this study?

Is there a difference between time 1 and time 2? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

76. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
77. the value of the descriptive statistic
78. the name of the inferential statistic (use the Acceptable Abbreviations Page)
79. the value of the inferential statistic
80. the p-value (**report exactly as it appears with no extra zeros and no rounding**)
81. This effect is best described as:
(A) not significant (B) significant but trivial (C) small (D) medium (E) large
82. What is the numerical value indicates the degree of association between scores at time 1 and scores at time 2?
83. The degree of association between scores at time 1 and scores at time 2 is best described as:
(A) not significant (B) significant but trivial (C) small (D) medium (E) large

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	h_sat_t1	31.78	76	4.33	.50
	h_sat_t2	30.28	76	4.83	

Results for husbands' satisfaction

t1 = time 1

t2 = time 2

Paired Samples Correlations

	N	Correlation	Sig.
Pair 1 h_sat_t1 & h_sat_t2	76	.68	.000

Paired Samples Test

	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Pair 1 h_sat_t1 - h_sat_t2	1.50	3.68	.42				75	

84. Compute the standard error of the mean for husbands at time 2.
85. Compute the pooled variance
86. Compute the lower end of the 95% confidence interval.
87. Compute the upper end of the 95% confidence interval.
88. At time 2, what is the distance that people's scores typically deviated from the group mean?
89. If this study were repeated over and over, how much would you typically expect the mean for men time 2 to deviate from the true population mean for married men after the birth of a child?
90. What is the average difference score (that is, the difference between a person's score at time 1 and his score at time 2)?
91. What is the typical distance that people's difference scores deviated from the group mean difference score?
92. If this study were repeated over and over, how much would you typically expect the size of group mean difference score to deviate from the true difference in the population?
93. What is the average deviation of $|e_{diff\ sp}|$?

Is there a difference between time 1 and time 2? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

94. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
95. the value of the descriptive statistic
96. the name of the inferential statistic (use the Acceptable Abbreviations Page)
97. the value of the inferential statistic
98. the p-value
99. This effect is best described as:
(A) not significant (B) significant but trivial (C) small (D) medium (E) large
100. What is the numerical value indicates the degree of association between scores at time 1 and scores at time 2?

Part E: More of “what’s it for?”

For each question below, you will be given some results from a particular analysis. You will need to identify what types of questions are addressed by the analysis, and what questions are not addressed. Answer each question with one of the following options: (A) yes, (B) no, or (C) the given information does not address this issue.

The output for the Levene’s test looks like this:

F	Sig.
5.363	.024

Based on this information:

101. Was the t-test for mean differences significant?
102. Is there a difference between means?
103. Is there a difference between standard deviations?
104. Were any assumptions violated?
105. Does the rank order of cases show stability across different assessment points?

The output for a paired sample correlation looks like this:

Correlation	Sig.
.318	.018

Based on this information:

106. Was the t-test for mean differences significant?
107. Is there a difference between means?
108. Is there a difference between standard deviations?
109. Were any assumptions violated?
110. Does the rank order of cases show stability across different assessment points?

Part F: predictive equations

WARNING: All your answers for questions in Part F should be entered in Canvas **as whole numbers with NO spaces and NO decimals**. Due to limitations with the Canvas grading system for these questions, Canvas is only able to process answers that are whole numbers, and if you type any extra characters (such as a space or period or letter) it will grade the response as incorrect.

111. Imagine you are conducting a study with a single sample to estimate a single mean and you get the dataset depicted below.

x
24
22
19
20
15

Write the linear equations for **case 1** assuming that the null hypothesis is **true**. To write these equations, simply replace each of the letters in the equations below with numerical values. (Assume the equations are listed in the same order below they are listed in class notes).

$$A = B + C$$

$$D = E + F$$

112. Imagine you are conducting a study with a paired sample to estimate a mean difference score and you get the dataset depicted below.

x_1	x_2	$x_1 - x_2$
24	18	
22	18	
19	17	
20	14	
15	13	

Write the linear equations for **case 1** assuming that the null hypothesis is true. To write these equations, simply replace each of the letters in the equations below with numerical values. (Assume the equations are listed in the same order below they are listed in class notes).

$$G = H + I$$

$$J = K + L$$

113. Imagine you are conducting an independent samples test for a difference between two means and you get the dataset depicted below.

y	x
19	1
17	1
13	1
11	1
13	0
12	0
13	0
10	0

Write the linear equations for **case 1 (the first case)** and for **case 8 (the last case)** assuming that the null hypothesis is true. To write these equations, simply replace each of the letters in the equations below with numerical values. (Assume the equations are listed in the same order below they are listed in class notes).

Case 1

$$A = B + (C) (D) + E$$

$$F = G + H$$

$$I = J + K$$

Case 8

$$L = M + (N) (O) + P$$

$$Q = R + S$$

$$T = U + V$$

Part G: Identify two things

Identify two answers that match each of the questions below.

114. What are two assumptions of an independent samples t-test?
 115. What are two different and unrelated types of effects that can be tested in a paired samples analysis?
 116. What are two things that increase the size of the standard error?

- A. A large standard deviation
 B. Difference between means
 C. A small sample

- D. Equality of variances
 E. Correlation between pairs
 F. Independence of observations

Part H: Written summaries

Paired samples summary

A sample of people completed a program for cantankerous individuals. The purpose of the program was to decrease participant's levels of rudeness. Each participant completed a Rudeness Scale at two points in time: once before completing the program (Time A) and once after completing the program (Time B). Results from this study are printed on the next page.

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	T1_rude	18.13	53	6.75	.93
	T2_rude	12.98	53	6.43	.88

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	T1_rude & T2_rude	53	.460	<.001

Paired Samples Test

		Paired Differences							
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
					Lower	Upper			
Pair 1	T1_rude - T2_rude	5.15	11.25	1.55	2.05	8.25	3.33	52	.002

117. **Write a summary of these results.** Your written summary should follow the template provided in your notes and described in the lecture add-on video. After you have provided a written summary, use the scoring system in the box below to grade your own answer. On Canvas, you will be asked to report if you wrote this summary and graded your own answer. The scoring system in the box below is the same as the scoring system that will be used to grade written summaries on learning assessments.

Scoring for a written summary of a paired samples analysis

This question is worth 9 points. Give yourself 1 point for correctly including each of the following components in your summary.

1. The first sentence needs to say there was a difference between the two time points. This needs to use the words “difference between” and needs to describe it as a difference between the two time points (e.g., between time 1 and 2, or between before and after the program). It should NOT describe it as a difference between two things that are not being compared in this study (for example, it is not a difference between groups, and not a difference between receiving the program and rudeness).
2. Summary reports: $d = .78$
3. Summary reports: $t(52) = 3.33$
4. Summary reports: $p = .002$
5. Summary states that rudeness was higher before the program and/or lower afterward.
6. Summary lists the mean and standard deviation for each time point, and these values appear at a point that is after the word “rudeness” has been used (in other words, after the reader has been given the name of the variable).
7. Summary reports that the effect is “medium.”

8. The summary is easy to read. Everything is in the order listed above or in some other sensible order. It does not contain unnecessary information. It does not contain sentences that are difficult to understand.
9. The summary includes information listed for d , and t , and it includes a statement about the magnitude of effect (trivial, small, medium, large). This point is awarded just for knowing you need to include these three crucial components regardless of whether the actual values are correct.

Deduct one-half point if a p-value is listed as “ $p=.000$.” Any time a p-value is less than .001 (or listed as “.000” in computer output), it should be printed as “ $p < .001$ ” in the text.

Deduct **3 points** if there is a major misinterpretation of the results, such as a mix up about whether results are significant, or a contradictory statement implying that results are both significant and not significant. In this case, the difference between timepoints was significant, and it would be a major misinterpretation to call this difference non-significant.

Below is an example of an answer that would receive full credit:

There was a significant difference between scores at Time 1 and Time 2 ($d = .78$, $t(52) = 3.33$, $p = .002$), with the people reporting higher rudeness before the program ($M = 18.13$, $SD = 6.75$) than after the program ($M = 12.98$, $SD = 6.43$). This represents a medium effect.

Independent samples summary

A study was conducted testing the extent to which physical exercise can increase feelings of emotional wellbeing. College students were randomly assigned either to a treatment condition completing a 3-week aerobic exercise program (group 1) or to a control condition (group 0). Afterward, they completed the Everyday Wellbeing Appraisal Scale (EWAS) to assess their level of emotional wellbeing. Output is depicted below.

Group Statistics

	Group	N	Mean	Std. Deviation	Std. Error Mean
EWAS	1	47	26.38	4.93	.72
	0	58	22.47	4.77	.63

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference		Lower	Upper
EWAS	Equal variances assumed	.091	.764	4.120	103	.000	3.92	.95		2.03	5.80
	Equal variances not assumed			4.106	97.151	.000	3.92	.95		2.02	5.81

118. **Write a summary of these results.** Your written summary should follow the template provided in your notes and described in the lecture add-on video. After you have provided a written summary, use the scoring system in the box below to grade your own answer. On Canvas, you will be asked to report if you wrote this summary and graded your own answer. The scoring system in the box below is the same as the scoring system that will be used to grade written summaries on learning assessments.

Scoring for a written summary of an independent samples analysis

This question is worth 9 points. Give yourself 1 point for correctly including each of the following components in your summary.

1. The first sentence needs to say there was a difference between the two groups. This needs to use the words “difference between” and needs to describe it as a difference between the two groups (e.g., between treatment and control, or between the exercise group and the no-exercise group). It should NOT describe it as a difference between two things that are not being compared in this study (for example, it is not a difference between exercise and emotional wellbeing).
2. Summary reports: $d = .81$
3. Summary reports: $t(103) = 4.12$
4. Summary reports: $p = < .001$
5. Summary states that people in the treatment/exercise condition had higher wellbeing (or emotional wellbeing, or everyday wellbeing appraisal) than people in the control condition.
6. Summary lists the mean and standard deviation for each group, and these values appear at a point that is after the word “wellbeing” has been used (in other words, after the reader has been given the name of the variable).
7. Summary reports that the effect is “large.”
8. The summary is easy to read. Everything is in the order listed above or in some other sensible order. It does not contain unnecessary information. It does not contain sentences that are difficult to understand.
9. The summary includes information listed for d , and t , and it includes a statement about the magnitude of effect (trivial, small, medium, large). This point is awarded just for knowing you need to include these three crucial components regardless of whether the actual values are correct.

Deduct one-half point if a p-value is listed as “ $p = .000$.” Any time a p-value is less than .001 (or listed as “.000” in computer output), it should be printed as “ $p < .001$ ” in the text.

Deduct **3 points** if there is a major misinterpretation of the results, such as a mix up about whether results are significant, or a contradictory statement implying that results are both significant and not significant. In this case, the difference between groups was significant, and it would be a major misinterpretation to this difference non-significant.

Below is an example of an answer that would receive full credit:

There was a significant difference between people in the exercise group and people in the control condition ($d = .81$, $t(103) = 4.12$, $p < .001$), with the people in the exercise group reporting more emotional wellbeing ($M = 26.38$, $SD = 4.93$) than people in the control condition ($M = 22.47$, $SD = 4.77$). This represents a large effect.

References

Note: Homework examples are designed to produce results similar to actual studies; however, details of studies have been modified to meet the needs of the assignment.

- Arigo, D., & Smyth, J. (2012). The benefits of expressive writing on sleep difficulty and appearance concerns for college women. *Psychology & Health*, 27(2), 210-226.
- Feeney, J., Alexander, R., Noller, P., & Hohaus, L. (2003). Attachment insecurity, depression, and the transition to parenthood. *Personal Relationships*, 10, 475-493.
- Sloan, D. M., Marx, B. P., Epstein, E. M., & Dobbs, J. L. (2008). Expressive writing buffers against maladaptive rumination. *Emotion*, 8(2), 302-306.